Noise Analisys in a Fiber Bragg Grating Accelerometer using Allan Variance Method

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ABSTRACT

我们利用艾伦方差法确定了Fiber Bragg光栅加速度计（FBGA）在静态运行时的噪声系数。我们描述了FBGA的机械结构，以及用于获取实验数据的嵌入式光学和电子电路。将我们拓扑的根艾伦方差图与两张MEMS加速度计图（一张来自理论文章，另一张来自商业数据手册）进行比较，我们观察到FBGA结果显示出相对较低的噪声剖面。

Keywords: Noise Coefficients, Allan Variance, Fiber Bragg Grating Accelerometer

1. INTRODUCTION

对传感器进行充分的噪声特性描述对于充分比较不同传感器拓扑结构至关重要。艾伦方差法是首选工具，因为它能提供有意义且直观的结果。艾伦方差法提供了均方根随机漂移误差作为平均时间函数的函数。它利用测量的数据点和采样率，对固定时间长度T的后续点簇进行时间平均。这些平均值用于计算艾伦方差σ2（T）。噪声系数随后通过艾伦方差计算，详见第3节。

本文中我们描述了Fiber Bragg光栅加速度计（FBGA）的一些关键噪声：偏置不稳定性、漂移率斜坡、量子化、速度随机游走、加速度随机游走和正弦波噪声。为了完整描述问题，先描述FBGA，接着是艾伦方法的数学描述和获得的噪声系数。

2. FIBER BRAGG GRATTING ACCELEROMETER DESCRIPTION

本研究中使用的FBGA由仅由光纤维持的地震质量组成，如图1所示。这是一种简化的表示方式，省略了用于固定光纤的安装墙。该设备的详细描述见其他地方2。使用了六根光纤，每根光纤内嵌有一根FBG。它们穿过地震体的面，并被预先拉伸，以确保感受到应变。该装置能够仅用一个地震质量测量三个轴的加速度。

为该应用开发了专用的光学询问器，如图2所示。在无加速度下，两侧FBG在同一轴的反射光谱存在部分重叠，重叠会根据加速度作用于地震质量所引起的变形而增减。耦合器将FBG1反射光的一部分分成一部分，提供其反射光谱部分（耦合器，输出4）强度的参考。耦合器通过光纤2产生的光源用于照明FBG2。任一FBGs的相似性决定了反射光的强度，反射光通过耦合器和光循环器返回到光电探测器PD2。

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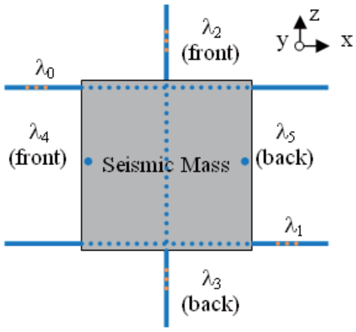

Figure 1. Accelerometer seismic mass (gray) sustained by optical fibers (blue).

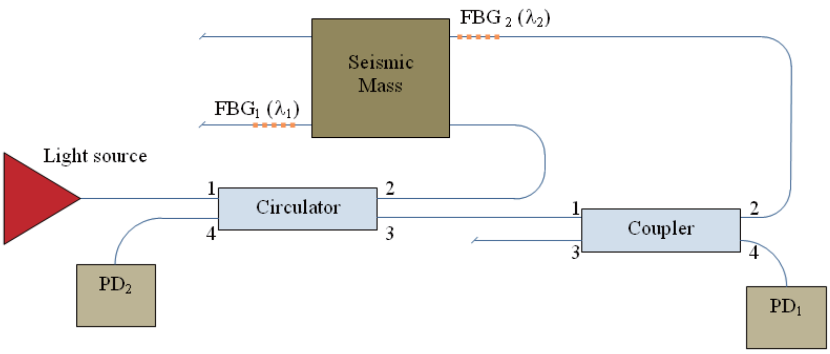

Figure 2. Embedded optical interrogator improved for ratiometric measurements.

已为模拟转数字（A/D）转换器输入，实现了PD1和PD2的足够信号（见图3）。比率转换以每秒100个采样率进行，组装成数据帧，发送并存储在标准计算机中。

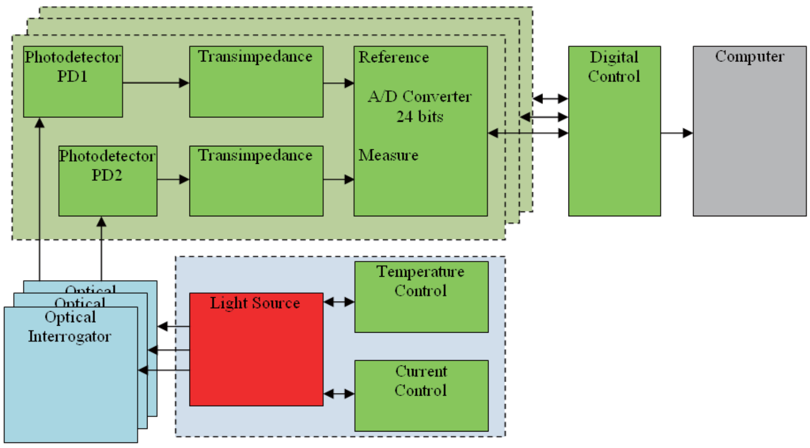

Figure 3. Embedded electronic used for data acquisition.

本次分析所用数据是在一个实验中收集的，加速度计在便携式实验室组装，实验室的积分盒放置在具有地震隔离和温度控制的桌子上。

3. ALLAN VARIANCE METHOD

本节详细介绍了如何通过艾伦方差（Allan Variance）方法获取噪声系数。
艾伦方差法的输入是收集的数据及其采样率。在我们的实验中，数据是加速度计静态作的测量。部分数据在图4中以示意方式表示。采样率为每秒1/τ点，T是n个点簇的周期，tk是与阵列第k点相关的时间。将 N 定义为数组中的点数，n 必须位于方程（1）所示区间内。

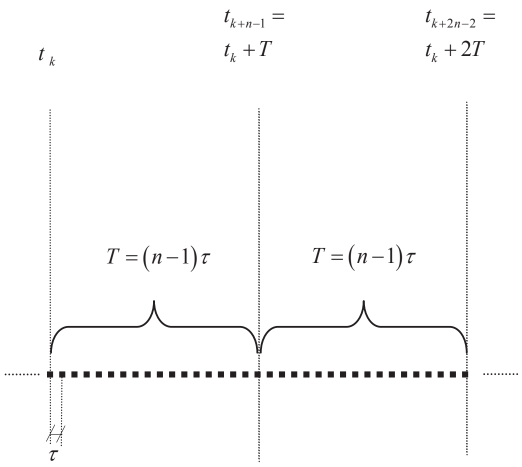

Figure 4. Data points array partial schematic representation.

$$2 \le n \le \frac{N+1}{2} \tag{1}$$

Let the output of the accelerometer as a function of time be denoted as $A(t)$. Then, the time average for each cluster of length T, $\overline{A}_k(T)$, is defined[1] according to Eq. (2):

$$\overline{A}_k(T) = \frac{1}{T}\int_{t_k}^{t_k+T} A(t)\,dt \tag{2}$$

Remembering that the data is discrete, Eq. (2) is rewritten using the left-hand Riemann sum as shown in Eq. (3):

$$\overline{A}_k(T) = \frac{1}{n-1}\sum_{p=k}^{k+n-2} A(t_p) \tag{3}$$

The Allan Variance of length T is defined as[1]:

$$\sigma^2(T) = \frac{1}{2}\left\langle \left[\overline{A}_{k+(n-1)}(T) - \overline{A}_k(T) \right]^2 \right\rangle \tag{4}$$

It follows from Eq. (3) and Eq. (4):

$$\sigma^2(T)=\frac{\sum_{p=1}^{N-2n+2}\left[\overline{A}_{p+n-1}(T)-\overline{A}_p(T)\right]^2}{2(N-2n+2)} \tag{5}$$

Let $S_{noise_i}(f)$ be the power spectral density of a specific type of noise (here denoted by $noise_i$). Considering that the random processes are all stationary in time, it follows that[1]:

$$\sigma^2_{noise_i}(T)=4\int_0^\infty S_{noise_i}(f)\frac{\sin^4(\pi fT)}{(\pi fT)^2}df \tag{6}$$

Then, the contribution of each noise to the Allan Variance as a function of T can be determined from the corresponding power spectral density. The Allan Variance for each noise[1] considered in this work is shown in Eqs. (7) to (12).

Bias Instability Noise

$$\sigma^2_{bias}(T)=\frac{2Q^2_{bias}}{\pi}\ln 2 \qquad for\ T>>\frac{1}{f_{cutoff}} \tag{7}$$

Drift Rate Ramp Noise

$$\sigma^2_{drift}(T)=\frac{Q^2_{drift}T^2}{2} \tag{8}$$

Quantization Noise

$$\sigma^2_{quant}(T)=\frac{3Q^2_{quant}}{T^2} \tag{9}$$

Velocity Random Walk Noise

$$\sigma^2_{velRW}(T)=\frac{Q^2_{velRW}}{T} \tag{10}$$

Acceleration Random Walk Noise

$$\sigma^2_{accRW}(T)=\frac{Q^2_{accRW}T}{3} \tag{11}$$

Sinusoidal Noise

$$\sigma^2_{\sin}(T)=Q^2_{\sin}\left[\frac{\sin^2(\pi f_0 T)}{\pi f_0 T}\right]^2 \tag{12}$$

The total noise contribution is given by Eq. (13).

$$\sigma^2_{AllNoise}(T)=\sum_i \sigma^2_{noise_i} \tag{13}$$

Where $f_{cuttoff}$ is the cutoff frequency of the bias instability noise, f_0 is the sinusoidal noise frequency, Q is the coefficient of each noise, σ is the root variance and the indices specifies each of the noises described.

其中fcuttoff是偏置不稳定性噪声的截止频率，f0是正弦噪声频率，Q是每个噪声的系数，σ是根方差，指标表示所描述的每种噪声。

4. EXPERIMENTAL RESULTS FOR NOISE COEFFICIENTS

The noise coefficients, denoted by Q_{noise_i} , are obtained fitting Eq. (13) into Eq. (5). The fitting obtained is shown in Figure 5. Table 1 presents the coefficients related to the fitting of Figure 5. In order to improve the fitting, two sinusoidal noise terms with different frequency f_0 (Eq. (12)) were used.

To evaluate the performance of the developed accelerometer, the Root Allan Variance plots obtained from two different MEMS accelerometers are used as reference. The first accelerometer is that contained in the ADIS16350 sensor[3]. The second accelerometer is that contained in the ADIS16385 sensor[4].

Comparing those two plots with that presented on Figure 5, the Fiber Bragg Grating Accelerometer analyzed in this work presents lower noise than those devices. Noise reducing techniques are being developed for future versions of this project to improve its noise characteristics, as temperature control of the whole system, modulation of the light source and synchronous detection and demodulation of photodetected signals. Increasing the power of light source may also reduce the noise coefficients; so, the use of erbium doped fiber optical sources may be adopted in future developments.

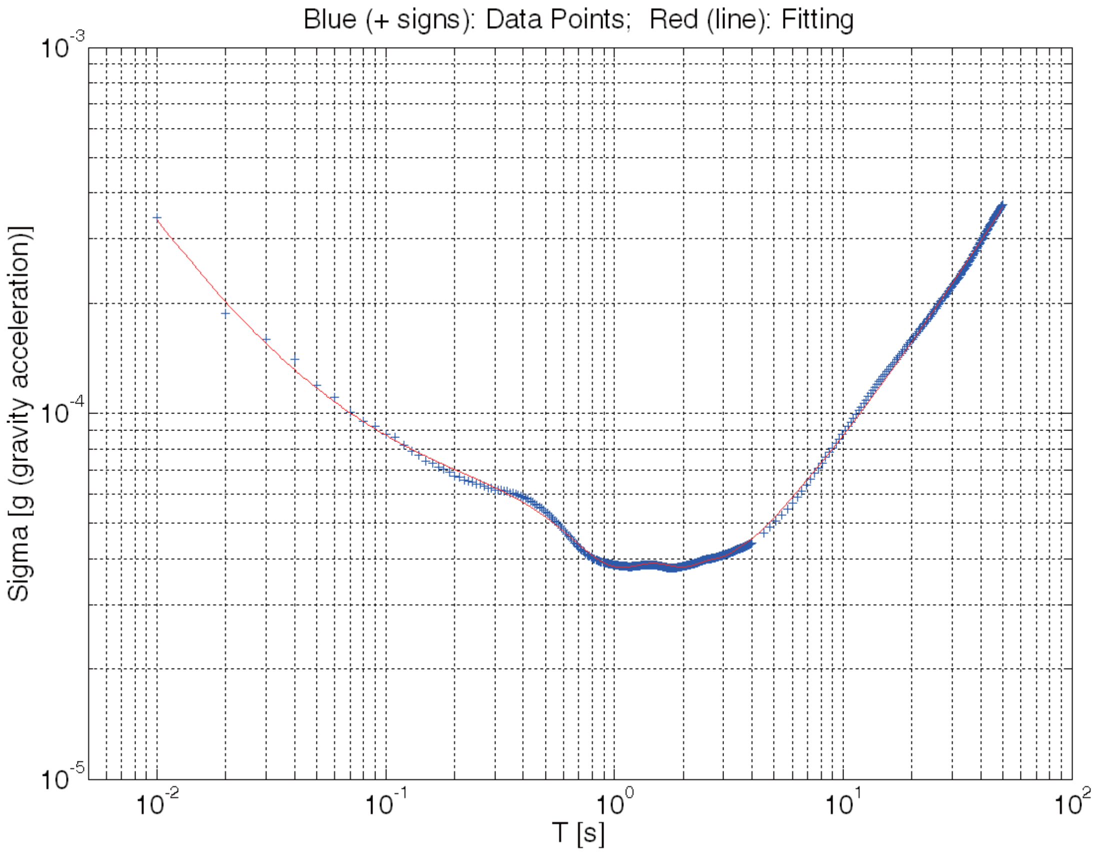

Figure 5 – Fitting obtained for noise coefficients estimation; dotted: Root Allan Variance plot; continuous: fitting using equations (7) to (12).

将这两张图与图5所示的图比较后，本研究分析的Fiber Bragg光栅加速度计的噪声低于这些设备。未来项目正在开发降噪技术，以改善其噪声特性，如全系统温度控制、光源调制以及光探信号的同步检测和解调。增加光源功率也可能降低噪声系数;因此，未来开发可能会采用掺铒光纤光源。为了评估已开发加速度计的性能，参考了从两种不同MEMS加速度计获得的Root Allan方差图。第一个加速度计是装在ADIS16350传感器里的。第二个加速度计是装在ADIS16385传感器4中。将这两张图与图5所示的图比较后，本研究分析的Fiber Bragg光栅加速度计的噪声低于这些设备。未来项目正在开发降噪技术，以改善其噪声特性，如全系统温度控制、光源调制以及光探信号的同步检测和解调。增加光源功率也可能降低噪声系数;因此，未来开发可能会采用掺铒光纤光源。

Table 1. Obtained Noise Coefficients with 95% confidence bounds.

Noise	Coefficient(s)*
Bias Instability	$Q_{bias} = (1.1 \pm 0.9).10^{-5}\ [g]$
Drift Rate Ramp	$Q_{drift} = (9.84 \pm 0.07).10^{-6}\ \left[\frac{g}{s}\right]$
Quantization	$Q_{quant} = (1.56 \pm 0.05).10^{-6}\ [g.s]$
Velocity Random Walk	$Q_{velRW} = (1.97 \pm 0.09).10^{-5}\ [g\sqrt{s}]$
Acceleration Random Walk	$Q_{accRW} = (2.8 \pm 0.2).10^{-5}\ \left[\frac{g}{\sqrt{s}}\right]$
Sinusoidal 1	$Q_{\sin_1} = (4.7 \pm 0.2).10^{-5}\ [g]$ $f_{0_1} = (0.96 \pm 0.04)\ [Hz]$
Sinusoidal 2	$Q_{\sin_2} = (2.7 \pm 0.3).10^{-5}\ [g]$ $f_{0_2} = (0.19 \pm 0.03)\ [Hz]$

* g: local gravity acceleration

5. CONCLUSIONS

The Allan Variance method was fully described and implemented in this article. Using this approach, seven noise sources were detected in the Fiber Bragg Grating Accelerometer described in section 3.

The curve fitting obtained (summarized in Figure 5 and Table 1) shows a high degree of accordance between the theoretical noise variance terms and the experimental data.

With those results, further analyses can be made, as scale linearity, temperature dependence and band pass, for example. Improvements of the system are possible, as modifications in the light source and signal processing techniques.

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本文对艾伦方差法进行了完整描述和实现。采用此方法，在第3节描述的Fiber Bragg光栅加速度计中检测到七个噪声源。获得的曲线拟合结果（见图5和表1总结）显示理论噪声方差项与实验数据高度一致。获得的曲线拟合结果（见图5和表1总结）显示理论噪声方差项与实验数据高度一致。